

# Installation Guide

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## Summary

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Upload the XMIT file `nfspd.v*.xmi` to MVS, to a dataset name of your choice and then "receive" the XMIT data into a PDS, for example `SYSS.NFSD.V0R2M0.DISTRIB`.

This will be a partitioned dataset, `RECFM=FB, LRECL=80` format that contains --

- Install JCL.
- A sample started task JES2 stored procedure.
- A sample configuration which you can place in its own dataset, or a member in `SYS*.PARMLIB`.
- XMIT format data to create the runtime load library.
- XMIT format data for the source datasets.

There are two executable programs which are used by the started task.

- **RESSOCK** - This program is similar to Jason Winter's **RESET** sample program. However, *RESSOCK* takes a list of port numbers for which sockets are to be reset. This program was very useful when NFSD was unstable and would crash leaving socket connections behind it. However, as NFSD (famous last words) is quite stable this probably isn't necessary at this time.
- **NFSD** - This is the executable for the NFS server as you might expect.

Note that the installation job does not copy the configuration file, or started task stored procedure. It is left for you to do this manually.

Note that the full source and development tree can be found on github at --

[https://github.com/twinslow/mvs\\_nfspd](https://github.com/twinslow/mvs_nfspd)

## Installation Steps

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### Step 1 - What HLQ are you going to use?

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In the examples for this installation document I have been using a HLQ of *SYSS*. This is not already in use on a standard *MVS TK5* installation.

You should avoid having the install datasets cataloged in the MVS master catalog, as this may result in the catalog entries being lost after maintenance.

There is a job provided in *MVS TK5* which defines all catalog aliases, pointing specific HLQ values to *USER catalogs*. This job is *SYS2.JCLLIB(ALLALIAS)*. I like to update that job with any new HLQs, as a record of what has been defined. The *IDCAMS* command is shown below to create the alias for *SYSS*.

```
DEF ALIAS(NAME('SYSS') REL('SYS1.UCAT.TK5'))
```

You can copy the *IDCAMS* command for the *define alias* and execute it directly via TSO option 6. Note that executing the command provides no output back to the screen, which is a little disconcerting.

You can verify that the command did indeed create the alias using the following command in option 6 (assuming you are using the *SYSS* HLQ) --

```
LISTC ENT('SYSS') ALL
```

You'll see output something like this ...

```
ALIAS ----- SYSS
IN-CAT --- SYS1.MCAT.TK5
HISTORY
RELEASE-----2
ASSOCIATIONS
USERCAT--SYS1.UCAT.TK5
***
```

## Step 2 - Create the DISTRIBution PDS from the XMIT file

There are a multitude of ways you can get the XMIT file onto your MVS system. If you are using *MVS TK5* I recommend using the full screen interface to *RECV370* provided as option *M.R*.

Option 2 will generate a batch job that will initialize the hercules reader device with the specified (PC or Linux) file and then execute the *RECV370* program to read the *card deck* directly from the reader unit device specified.

Below are screen shots for both options (you only need to use one of them).

In these instructions I am assuming this output dataset is -- **SYSS.NFSD.V0R2M0.DISTRIB** .

## If the XMIT file is NOT on MVS

This option submits a job that will --

1. Issue a DEVINIT command to hercules, to place the specified XMIT format (EBCDIC/BINARY) file on the specified reader device.
2. Run RECV370 to process the XMIT file and create the output dataset.
3. Issue another DEVINIT command to hercules, to reset the reader with no file loaded.

Note the use of forward slashes in the PC file name.

```
----- Receive XMIT File -----
OPTION  ==> 2

1  Receive XMIT file on MVS to PDS/SEQ file          HERC01
2  Receive XMIT file on PC  to PDS/SEQ file          PRECV372
                                                    PXMI

XMIT Input File
1 MVSFILE:
-or-
2 PCFILE: C:/mvs-clean/nfsd_v0r2m0/nfsd_v0r2m0.xmi
Reader Control for PCFILE
HercRDR Jes2RDR DEVINIT Reset Command
10C              DEVINIT 10C

MVS Output File
DSN: SYSS.NFSD.V0R2M0.DISTRIB                      VOL: TS0003  UNIT: SYSDA
SPACE: (CYL,(5,2,3))                               DISP: (NEW,CATLG)

JOB STATEMENT INFORMATION
==> //HERC01R JOB CLASS=A,
==> //          MSGLEVEL=1,MSGCLASS=X,NOTIFY=HERC01
==>
==>
```

## If the XMIT file is already on MVS

After using FTP or your terminal emulator (such as the wonderful Vista TN3270) to load the file to MVS, you can use *RECV370* to create the distribution dataset. **Note**, I had to create the dataset first for both FTPD and IND\$FILE transfer using Vista TN3270.

In the screen shot below, I've shown the selected options using TK5 full screen interface to *RECV370* that will submit a batch job and create the distribution dataset.

```

----- Receive XMIT File -----
OPTION ==> 1

1 Receive XMIT file on MVS to PDS/SEQ file          HERC01
2 Receive XMIT file on PC to PDS/SEQ file          PRECV372
                                                    PXMI

XMIT Input File
1 MVSFILE: SYSS.NFSD.V0R2M0.XMI
-or-
2 PCFILE:
  Reader Control for PCFILE
  HercRDR Jes2RDR DEVINIT Reset Command

MVS Output File
DSN: SYSS.NFSD.V0R2M0.DISTRIB                      VOL: TS0003  UNIT: SYSDA
SPACE: (CYL,(5,2,3))                               DISP: (NEW,CATLG)

JOB STATEMENT INFORMATION
==> //HERC01N JOB CLASS=A,
==> //      MSGLEVEL=1,MSGCLASS=X,NOTIFY=HERC01
==>
==>

```

## Step 3 - Create the NFSD source and load library datasets

Customize the install job `SYSS.NFSD.V0R2M0.DISTRIB(INSTALL)` . You will likely wish to change --

- Job card information such as job name, job class, message class, notify userid etc.
- The dataset name of your `DISTRIB` PDS.
- The target output dataset prefix for load library and source datasets.
- The target volume for these datasets.

Note that these datasets are created as `DISP=(NEW,CATLG)` and the job will fail if they already exist.

The NFSD server does not need to run authorized, but it will need to have appropriate permissions to its configuration and the datasets to be exported (the datasets that are to be remotely mounted and accessed).

In the *MVS TK5* environment, the started task will run under the `STC` userid, and have a group of `STCGROUP` . The permissions in the default setup should be suitable, allowing read/write access to any

dataset you wish to export. See *TK5* and *RAKF* documentation for more information on this. If you have RAKF access failures, check the settings in *SYS1.SECURE.CNTL(PROFILES)*.

## Step 4 - Customize, or create the started task stored procedure

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The member `SYSS.NFSD.V0R2M0.DISTRIB(NFSD)` is a sample stored procedure you can copy to a suitable JES2 procedure library. If you are using the MVS TK5 system then I suggest using `SYS2.PROCLIB`.

- Update the load library dataset as required.
- Uncomment and change SYSOUT class, or destination of SYSUDUMP as appropriate.
- Update the source of the configuration file (see below).

## Step 5 - Customize, or create the NFSD configuration file

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The member `SYSS.NFSD.V0R2M0.DISTRIB(CONFIG)` contains a sample configuration for the NFSD server. You can copy this to either a sequential dataset, or a member of a PDS. The dataset can be record format FB or VB. You may wish to keep the configuration as a member of `SYS1.PARMLIB`, such as `NFSDCFG0`.

The sample JES2 stored procedure assume exactly this setup.

Note that at this time the dataset names in the export list cannot include wildcards to perform generic exports based on system catalog. You must explicitly list each dataset you wish to export.

## Step 6 - Start NFSD

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Enter a start command on the MVS console to run NFSD, `S NFSD`.

# MVS Modify Commands

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The logging level and options can be changed via a MVS modify command.

For example to set the level of messages being output to the log use the command --

```
F NFSD,SET LOGLVL INFO
```

The log levels (INFO is show above) are --

- DEBUG

- TRACE
- INFO
- WARN
- ERROR
- FATAL

You can separately set the level of messages that are output as WTO messages to the console.

```
F NFSD,SET WTOLVL WARN
```

The above command will cause NFSD to output only WARN messages and worse, to the MVS console.

You can also independently set logging level for different NFS operations.

```
F NFSD,SET LOGLVL DEBUG PROC=WRITE
```

This would cause any NFS WRITE operation to output DEBUG level messages, overriding (for example) a global INFO logging level.

The NFS operations that can be logged using the `PROC=` keyword are --

- GETATTR
- SETATTR
- LOOKUP
- ACCESS
- READ
- WRITE
- CREATE
- REMOVE
- RENAME
- READDIR
- READDIRPLUS (or the alias of RDIRPLUS)
- FSSTAT
- FSINFO
- PATHCONF
- COMMIT
- NULL

# Shutting down NFSD

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Issue an MVS STOP command to shutdown NFSD --

```
P NFSD
```